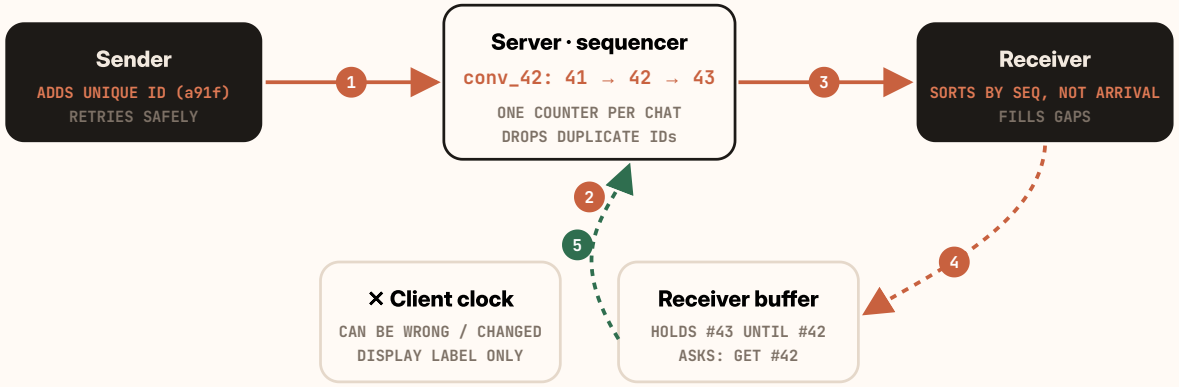
• SYSTEM DESIGN UNLOCKED · EP 05

Design WhatsApp: *Message Order*

Why "I'm running late" never shows up as "late I'm running", and why no message appears twice.

THE FLOW



1 Sender attaches a unique ID it generated (idempotency key)

2 Server stamps the next number from that chat's counter

3 Receiver shows messages by sequence number, not arrival

4 #43 before #42? Hold #43 in a buffer

5 Ask the server for the missing #42, then show both

× Never order by client time: it's just a display label

WHY ORDER BREAKS

Different routes. Each message travels separately; one path is slower.

Retries. A lost message is re-sent and lands after newer ones.

Clocks lie. Phones can be set to any time; even servers drift by milliseconds.

REQUIREMENTS

- Same order on every device in the chat
- Order follows the server, not device clocks
- No message shown twice, even after retries
- Missing messages detected and fetched
- Scales to huge groups without one global bottleneck

1	1×	0
COUNTER PER CHAT	STORED, HOWEVER MANY RETRIES	CLOCKS USED FOR ORDERING

EDGE CASES → FIX

Phone clock set to 2019 → ignored for ordering; server sequence decides.

#43 arrives before #42 → buffer #43, request #42, render both in order.

Same message retried 5× → server dedupes on the client-generated ID; stored once.

Two group members send at once → first to reach the sequencer gets the lower number; everyone sees the same order.

Hot, huge group → counter per chat (never global); shard chats across sequencers.

TRADE-OFFS TO SAY OUT LOUD

Sequencer vs timestamps: exact order, but one counter per chat must be fast and highly available.

Per-chat vs global counter: scales horizontally; there's no order across different chats (and none is needed).

At-least-once + dedupe: nothing lost, but every message needs a unique ID.

Alternatives: Lamport clocks / Snowflake-style IDs avoid a central counter but give weaker or approximate order.

INTERVIEW KEYWORDS TO SAY

`sequence number` `per-conversation counter` `sequencer` `clock skew` `client timestamp = display only` `idempotency key` `deduplication` `at-least-once delivery` `gap detection` `Lamport clock` `Snowflake ID` `hotspot`

Sources: Lamport, "Time, Clocks, and the Ordering of Events in a Distributed System" (1978) · techinterview.org (chat ordering) · system design references on per-conversation sequencing. This is the standard interview design, not a claim about WhatsApp's internal code.